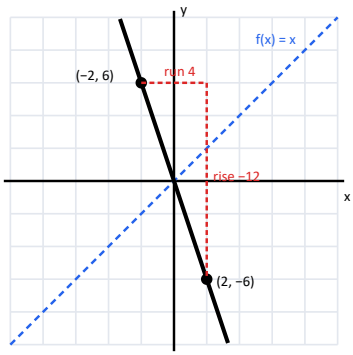
Exit Ticket Answer Key

Module 2, Topic 2, Lesson 1: Transforming Linear Functions

1. Given f(x) = x and m(x) = a · f(x), what is the value of a?



a = −3

1. m(x) = a · x, so a is the slope of m.
2. Slope from (−2, 6) to (2, −6): (−6 − 6) / (2 − (−2)) = −12 / 4 = **−3**.
   Or count on the grid: from (−2, 6) go right 4 and down 12, so slope = −12/4 = −3.
3. Check: m(2) = −3 · 2 = −6 ✓
4. Meaning: f(x) = x is stretched vertically by 3 **and** reflected over the x-axis (the negative sign).

> **If they wrote 3:** they missed the reflection. The line falls left to right, so a must be negative.
> **If they wrote −1/3:** they found run over rise. Slope is rise over run.

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2. How are the graphs of f(x) and g(x) = f(x) − 4 related?

A. translated up 4 units

B. translated down 4 units ✓

C. translated right 4 units

D. translated left 4 units

1. The −4 is **outside** the parentheses, so it changes the **outputs** (y-values).
2. Every y-value goes down 4: the point (x, y) moves to (x, y − 4). This is f(x) + d with d = −4.

> **If they chose A:** they ignored the sign. Minus outside means down.
> **If they chose C or D:** they mixed outside with inside. f(x − 4) would be right 4 (that is review Q4). Send them to the Lesson 2 video.

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3. What is the y-intercept of g(x) = 2 · f(x)?

x	f(x)	g(x)
−1	−1	**−2**
0	2	**4**
1	5	**10**
2	8	**16**

y-intercept = (0, 4)

1. The y-intercept is where x = 0. From the table, f(0) = 2.
2. g(0) = 2 · f(0) = 2 · 2 = **4**.
3. A vertical stretch multiplies **every** output by 2, including the y-intercept. The slope doubles too: f(x) = 3x + 2, so g(x) = 6x + 4.

> **If they wrote 2:** they read f(0) and forgot to multiply by 2.
> **If they wrote 16:** they doubled f(2) = 8. The y-intercept always uses the x = 0 row.

Bonus (Lesson 2 video): f(x) = x and g(x) = f(x + 3). Which way and how far?

Left 3 units

1. The +3 is **inside** the parentheses, so it changes the **inputs** and moves the graph left or right.
2. Inside works the opposite of how it looks: f(x + 3) moves **left** 3. Example: (0, 0) moves to (−3, 0).
3. Also accept "up 3": g(x) = x + 3, and for f(x) = x only, left 3 and up 3 give the same line.

Use the results tomorrow: Missed Q1 → reteach slope sign and reflection (Lesson 1 video, "−f" part). Missed Q2 or Bonus → inside vs. outside (Lesson 2 video). Missed Q3 → "stretch multiplies every output" with a table. All correct → ready for Lesson 3.